

Position tracking control of a permanent magnet steering wheel equipped to a vertical tank robot: From theory, simulation to practice

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Abstract

In the domain of mobile robots (MRs) utilizing multi-wheel directional control, position synchronization in steering wheels remains a challenging problem, particularly when implemented in permanent magnet steering wheels for vertical tank robots. Addressing this critical problem, this paper presents a position synchronization control strategy that integrates a mean deviation coupling strategy (MDCS) with state feedback tracking control (SFTC) enhanced by integral action. In essence, MDCS minimizes position synchronization errors, especially during deviations or fluctuations, while SFTC enables fast responses to state changes and improves adaptability in uneven terrain. Combined with integral action, SFTC compensates for static errors, enhancing long-term stability. Furthermore, to establish robust evidence for the proposed strategy, this paper validates the results through both simulation and experimental analysis. Initially, simulation results show that integrating the MDCS with SFTC improves settling time, reducing it from 0.6 to 0.4 s. Additionally, experimental tests demonstrated that the improvement in settling time closely matched the simulation results during the reference angle change phase, excluding the initial startup phase. Without the synchronization strategy, the settling time was 0.5 s, whereas applying the control strategy increased it to 0.7 s. This discrepancy arises from the selected MDCS control coefficients, which were not well-suited to the actual system, leading to ineffective compensation signal intervention. As a result, the system dynamics were altered, and the stability of the poles was no longer ensured compared to the original selection. It can be concluded that the integration of MDCS with SFTC holds strong potential for practical application in steering wheels, improving synchronization while reducing the computational burden on hardware.

Keywords

mobile robots, permanent magnet steering wheel, mean deviation coupling strategy, state feedback tracking control, integral action

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Introduction

In the realm of cleaning and defect inspection, the vertical tank robot (VTR) is a promising candidate,^{1–3} offering exceptional flexibility for various applications of vertical terrain, comprising tank inspections, high-rise building cleaning, and hull cleaning.^{4–7} However, a significant challenge for VTR actuators arises from the substantial force required for adhesion to the working surface. For VTRs performing these tasks, the robot's structure consists of rolling wheels and steering wheels, allowing the robot to navigate narrow working spaces. During operation, when the rolling wheels are engaged, the characteristics of the steering wheels become critical for navigation. Specifically, the frictional force resulting

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from adhesion at each wheel can introduce control lag, instability, or unintended oscillations in position control. Consequently, this results in a lack of synchronization in the steering angles between the wheels, posing challenges in accurately controlling the heading angle for VTRs.

Reviewing previous studies, researchers have explored many approaches to address the challenges associated with the steering wheel dynamics in the domain of MRs. From a mechanical design perspective, Yuki Ueno et al.⁸ proposed a mechanical driving system to constrain the motion of the steering wheels. While this method can improve steering angle synchronization, it comes at the cost of increased mass and complexity, particularly when scaling up the number of steering wheels. In terms of improvement using control algorithms, Jiwei Qu et al. provided a motion mode switching algorithm to ensure smooth steering angle control and improve the angular position error between the wheels.⁹ Notwithstanding, this approach only considers scenarios without external forces acting on the steering wheels, limiting its applicability in real-world environments. A more viable study is in accordance with Chun-Yang Lan et al., who integrated a solution that combines nonsingular fast terminal sliding mode control with improved deviation coupling control to enhance disturbance rejection in motor position synchronization.¹⁰ Nonetheless, this approach demonstrated improved performance in maintaining steering angle synchronization, but it also comes with the drawback of high computational complexity, which can pose challenges as the number of motors increases. Consequently, these studies highlight ongoing efforts to develop effective solutions for steering wheel dynamics in MRs. As the complexity of these systems continues to grow, there is a demand for innovative design approaches and control algorithms that can provide reliable and efficient steering control, while also considering the practical constraints of cost, weight, and computational resources.

Accordingly, the state feedback tracking control (SFTC) presents a promising avenue for addressing the steering wheel synchronization challenges in MRs. Specifically, the SFTC offers a relatively simple structure and reduced computational,^{11–13} enabled by its discrete-domain operation,¹⁴ making it an attractive option for practical implementation. Nevertheless, a key limitation of SFTC is its inability to self-correct steady-state errors when the control system is subjected to long-term disturbances, changing operating conditions, or inaccurate models.¹⁵ To overcome this, SFTC is combined with integral action to enhance its ability to compensate for these issues.^{16,17} When applying SFTC in synchronization control strategies for MRs, the challenge lies in the fact that the controller relies solely on system state feedback, rather than deviation-based compensation. This limitation poses difficulties in simultaneously regulating the steering angles of multiple wheels, a critical requirement for maintaining accurate heading control. To address this challenge, the

integration of the mean deviation coupling strategy (MDCS) with SFTC is considered a suitable approach. The MDCS intervenes directly on the output value of the controller,¹⁸ complementing the state feedback-based control and enabling more effective synchronization of the steering angles across the multiple wheels. This combined strategy has the potential to enhance the overall performance and reliability of MRs in cleaning and inspection tasks.

To address these gaps, this study proposes a novel hybrid control strategy that combines SFTC with integral action and integrates it with MDCS to enhance the synchronization of permanent magnet steering wheels in VTRs. The proposed approach is developed based on a complete theoretical framework. Importantly, simulation results are provided to validate the correctness and performance of the control scheme at each stage of the design process. Additionally, the VTR model is implemented to experimentally assess the behavior and efficiency of the proposed algorithm for synchronous steering angle control. The main contributions of this paper are summarized as follows:

- A hybrid control strategy (SFTC + integral action + MDCS) is developed to address the synchronization problem of steering wheels in VTRs, balancing control performance and computational feasibility.
- The control design is built upon a discrete-time framework, enabling straightforward implementation on embedded hardware platforms.
- The effectiveness of the proposed strategy is validated through both simulation and experimental evaluation, demonstrating its potential for real-world application in mobile robot systems with multiple actuators.

System identification

Problem statement

The SFTC design is founded on the system's input-output response relationship,^{19,20} commonly formulated using the autoregressive exogenous (ARX) model. This indicates that the model's accuracy strongly affects the controller's quality. Therefore, it is necessary to obtain an ARX model for the control system before considering the SFTC design problem. There are two approaches to achieving a system model, including modeling using the first principles in engineering (white box) and experimental-based identification (gray box or black box).²¹ However, the white box method lacks efficiency due to the complexity of accurately determining system parameters.²² Additionally, external influences such as disturbance in the control system also cause significant deviations in the model. Therefore, it is not feasible to rely on the physical parameters of the system to determine the model and design the controller. According to the aforementioned issues, this study

approaches the identification of system models based on experimental data, specifically utilizing the recursive least squares (RLS) algorithm. Notably, the RLS algorithm requires prior knowledge of the form of the model, including the number of degrees of the autoregressive model polynomial and the exogenous input model. The initially chosen number of degrees can lead to different identification results; therefore, the RLS algorithm is considered a gray box-based method.

Model identification using RLS

The RLS identification method typically yields a single-input-single-output (SISO) ARX model described as follows:

$$A(q^{-1})y(k) = B(q^{-1})u(k), \quad (1)$$

where $y(k) \in \mathbb{R}$, and $u(k) \in \mathbb{R}$ are the system output and the system input at the k -th sampling, respectively; $A(q^{-1})$ and $B(q^{-1})$ are the rational polynomials of degrees n and m , representing the autoregressive model, and the exogenous input model, respectively, given by:

$$A(q^{-1}) = 1 + \sum_{i=1}^n a_i q^{-i}, B(q^{-1}) = \sum_{i=1}^m b_i q^{-i}, \quad (2)$$

with a_i, b_i are the i -th polynomial coefficients, and q^{-i} is the i -th order backward shift operator, defined as $q^{-i}f(k) = f(k - i)$. In which, a_i are auto-regression parameters that represent the influence of previous values on the current value of the dependent variable (output signal $y(k)$). Meanwhile, b_i are exogenous parameters that represents the impact of the input variable $u(k)$ on the dependent variable. The objective is to design the control input $u(k)$ for governing the system output $y(k)$

tracking closely to the reference input $r(k)$ with desired characteristics.

Case study: System identification of a DC motor

To elucidate the step process and illustrate controller design, this section demonstrates the system identification of a DC motor using the RLS method. The simulation model of the DC motor is built as a multi-physics system model using Simscape, as shown in Figure 1. The simulation process is performed in the Simulink environment, which is a MATLAB-based graphical programming space commonly used in simulation and analysis applications.²³ In which, the black lines describe the signal flow containing the computational data in the Simulink environment. Meanwhile, the remaining lines in the Simscape environment include electrical signals (voltage, current, etc.) and mechanical signals (torque, position, etc.). A Simscape DC motor block is configured using the specifications of a Maxon 388985 DC motor, with parameters listed in Table 1. The motor load torque is arbitrarily set to a constant

Table 1. Setting parameters of Simscape DC motor block.

Setting	Value
Field type	Permanent magnet
Model parameterization	By stall torque and no-load speed
Armature inductance	0.0308 mH
Stall torque	15,700 mN.m
No-load speed	4070 rpm
Rated DC supply voltage	24 V
Rotor damping parameterization	By no-load current
No-load current	1030 mA
DC supply voltage when measuring no-load current	24 V

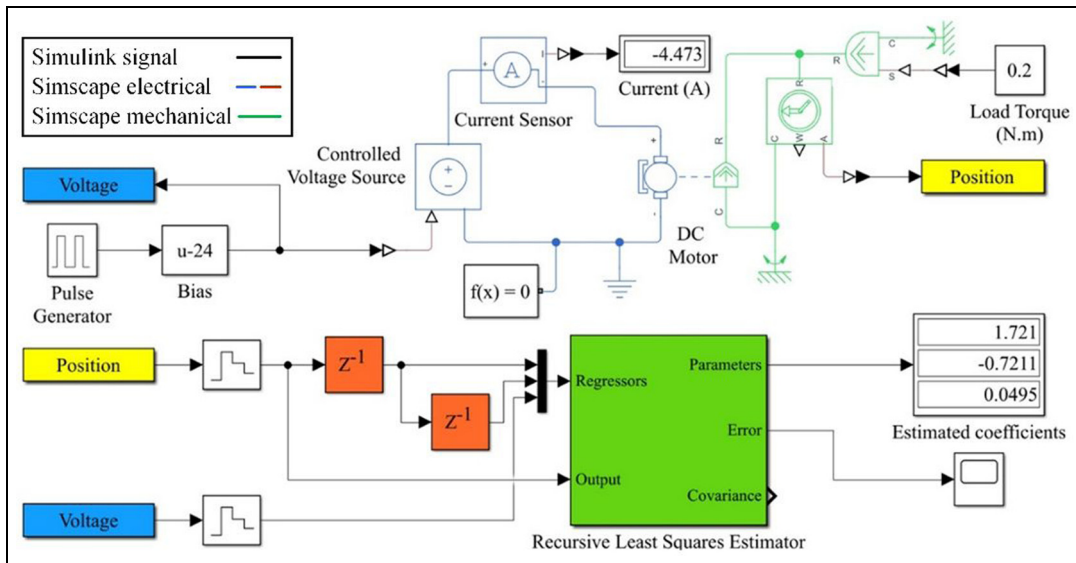


Figure 1. Schematic diagram of DC motor identification using MATLAB/Simulink with Simscape.

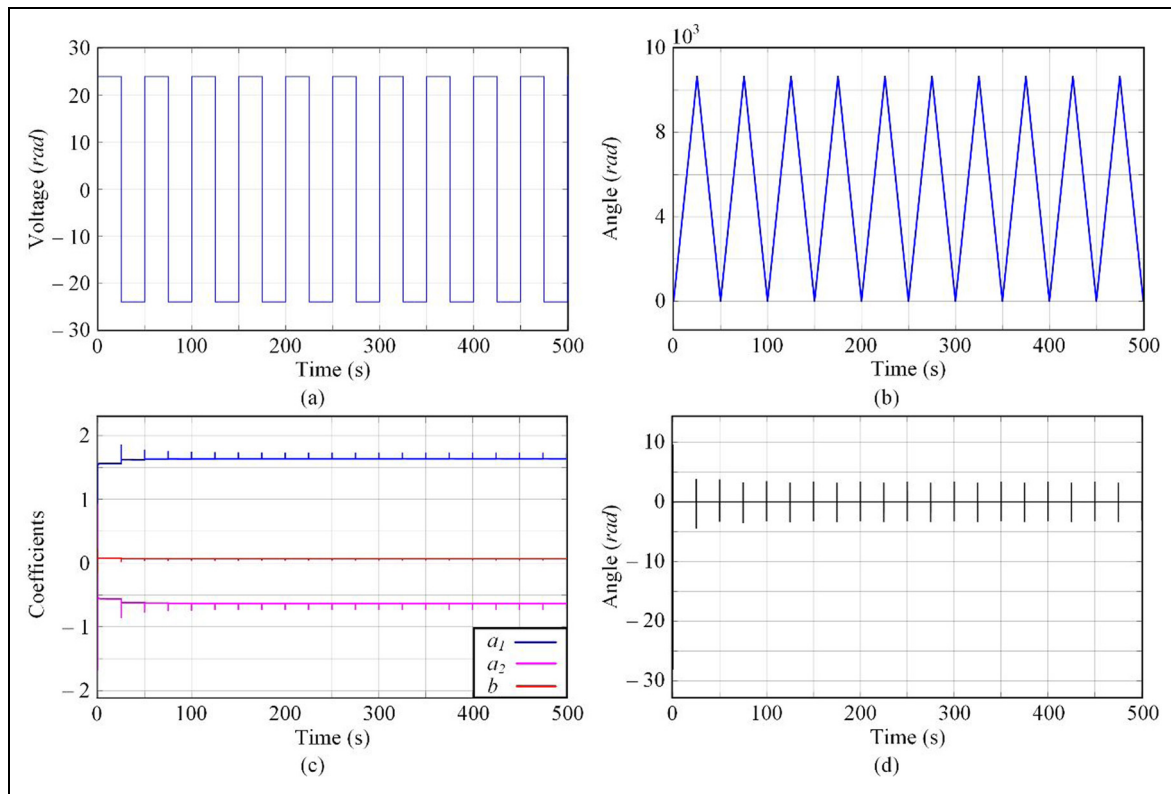


Figure 2. Result of model identification simulation for Simscape DC motor using RLS algorithm include (a) input signal, (b) output signal, model parameters (c), and estimated error (d).

0.2 N.m. It is important to note that these parameters of the DC motor are only intended for process validation and might not be readily available in practice. Since the objective control of the practical application is position tracking, the rotor angle (in radians) is fed to the RLS estimator through a ZOH block with a sampling time of 0.01 s. Additionally, the supply voltage for the DC motor periodically changes from -24 V to 24 V, modulated by a pulse generator and a bias with a frequency of 0.02 Hz. The RLS estimator identifies three parameters, including the rational polynomial $A(q^{-1})$ of degree 2 and the rational polynomial $B(q^{-1})$ of degree 0, determining the coefficients in sequence a_1, a_2 , and b_o . It should be noted that the aim of this paper is to propose an algorithm, and ARX model identification serves as an initial step to form the basis for design. Therefore, this study does not focus on optimizing the number of model degrees, and the choice of low degrees is for the purpose of computational simplicity. The simulation time is set to 500 s to ensure that the coefficients of the rational polynomials sufficiently capture the characteristics of the DC motor.

Figure 2 illustrates the signal flows and simulation results of the ARX model identification process for the Simscape DC motor using the RLS algorithm. The results show that the system coefficients after 25 s start to converge to a specific value. In addition, the estimation error increases abruptly at the input change times, even though the parameters have converged before.

This indicates that the identified model does not accurately represent the system's behavior during the initial period following the input change; however, it starts to match the system response after approximately 0.1 s. Based on Figure 2(c), the estimated coefficients are determined as $a_1 = 1.72, a_2 = -0.72$, and $b_o = 0.05$, respectively. In brief, the ARX model of the DC motor, given these specifications, is represented as

$$y(k) = 1.72y(k-1) - 0.72y(k-2) + 0.05u(k). \quad (3)$$

State feedback approach

State feedback tracking control

A position tracking control input using state feedback is given by

$$u_{sf}(k) = -K_{sf}(q^{-1})y(k) + G_{sf}(q^{-1})r(k), \quad (4)$$

where the subscript “sf” denotes elements related to the state feedback approach; $K_{sf}(q^{-1})$ and $G_{sf}(q^{-1})$ are polynomial control gains of degrees n_K and n_G , respectively, that are designed to ensure the closely tracking of the system output to the reference input.

By using the state feedback tracking control input of equation (4), the closed-loop control system of equation (1) can represent as

$$(B(q^{-1})K_{sf}(q^{-1}) + A(q^{-1}))y(k) = B(q^{-1})G_{sf}(q^{-1})r(k). \quad (5)$$

The characteristic polynomial of the closed-loop control system of equation (5) is defined as

$$\Delta_{sf}(q^{-1}) = B(q^{-1})K_{sf}(q^{-1}) + A(q^{-1}), \quad (6)$$

where $\Delta_{sf}(q^{-1})$ is the characteristic polynomial of degree n_Δ of the closed-loop control system using the state feedback control approach.

This characteristic polynomial matches the desired characteristic polynomial, which is represented in the following form

$$\Delta^d(q^{-1}) = (1 - p_1 q^{-1})(1 - p_2 q^{-1}) \dots (1 - p_{n_\Delta} q^{-1}), \quad (7)$$

where $p_1, p_2, \dots, p_{n_\Delta}$ are the desired poles in z -plane.

By equating the two characteristic polynomials and matching the coefficients of $q^{-1}, q^{-2}, \dots, q^{-n_\Delta}$, the polynomial control gain $K_{sf}(q^{-1})$ can be obtained. To find the polynomial control gain $G_{sf}(q^{-1})$, it is necessary to consider the system of equation (1) in the steady state, that is

$$A(q^{-1})y_{ss} = B(q^{-1})u_{ss}, \quad (8)$$

where y_{ss} and u_{ss} are respectively the system output and the control input at the steady state, i.e. $\lim_{k \rightarrow \infty} y(k) \rightarrow y_{ss}$, $\lim_{k \rightarrow \infty} u(k) \rightarrow u_{ss}$.

Let $\tilde{y}(k)$ be defined as $\tilde{y}(k) \triangleq y(k) - y_{ss}$, which represents difference between the system output at the sampling of k and the system output in the steady-state; $\tilde{u}_{sf}(k) \triangleq u_{sf}(k) - u_{ss}$ is the difference between the control input at the sampling of k and the control input in the steady-state. The objective of the control input is that the steady-state output y_{ss} tracks the reference $r(k)$, yielding $y_{ss} = r(k)$. Thus, the ARX model of equation (1) can be represented in the shifted domain, with $\tilde{y}(k)$ serving as the state variable,

$$\tilde{y}(k) = A(q^{-1})^{-1}B(q^{-1})u_{sf}(k) - y_{ss}. \quad (9)$$

Solving equation (8) for y_{ss} , and then substituting it into equation (9), it yields

$$\begin{aligned} \tilde{y}(k) &= A(q^{-1})^{-1}B(q^{-1})u_{sf}(k) - A(q^{-1})^{-1}B(q^{-1})u_{ss} \\ &= (A(q^{-1})^{-1}B(q^{-1}))\tilde{u}_{sf}(k). \end{aligned} \quad (10)$$

A state feedback control input can drive the state variable $\tilde{y}(k)$ in shifted domain to zero, i.e. $\tilde{y}(k) \rightarrow 0$, which implies that $y(k) \rightarrow r(k)$ and results

$$\tilde{u}_{sf}(k) = -K_{sf}(q^{-1})\tilde{y}(k). \quad (11)$$

The tracking control problem is thus converted into a simple regulator problem. Referring to $\tilde{u}_{sf}(k) = u_{sf}(k) - u_{ss}$ and using equation (11), the stable state feedback control input is rewritten as

$$u_{sf}(k) = -K_{sf}(q^{-1})y(k) + K_{sf}(q^{-1})y_{ss} + u_{ss}. \quad (12)$$

Comparing to the position tracking control input of equation (4), it yields the following constraint

$$K_{sf}(q^{-1})y_{ss} + u_{ss} = G_{sf}(q^{-1})r(k). \quad (13)$$

On the other hand, recalling that $A(q^{-1})y_{ss} = B(q^{-1})u_{ss}$, it can be represented as

$$\begin{aligned} &-(B(q^{-1})K_{sf}(q^{-1}) + A(q^{-1}))y_{ss} + B(q^{-1}) \\ &(K_{sf}(q^{-1})y_{ss} + u_{ss}) = 0. \end{aligned} \quad (14)$$

Since the steady state system output y_{ss} towards the reference input $r(k)$, it can infer from equations (13) and (14) that

$$G_{sf}(q^{-1}) = (B(q^{-1})K_{sf}(q^{-1}) + A(q^{-1}))B(q^{-1})^{-1}. \quad (15)$$

In summary, the system output $y(k)$ tracking to the reference input $r(k)$ under the control of the position tracking control input from equation (4). The polynomial control gain $K_{sf}(q^{-1})$ is determined by equating and matching the coefficients of the two characteristic polynomials, while the polynomial control gain $G_{sf}(q^{-1})$ is determined by equation (15).

Example of state feedback tracking control design

For a sake of illustration, the ARX model identified from the DC motor described in Section ‘Model Identification using RLS’ is employed to design the state feedback tracking control. From equation (3), the autoregressive model and the exogenous input are governed by

$$A(q^{-1}) = 1 - 1.72q^{-1} + 0.72q^{-2}, \quad B(q^{-1}) = 0.05. \quad (16)$$

Consequently, the characteristic polynomial of equation (6) with $K_{sf}(q^{-1}) = k_1 q^{-1} + k_2 q^{-2}$ is derived as

$$\Delta_{sf}(q^{-1}) = 1 + (0.05k_1 - 1.72)q^{-1} + (0.05k_2 + 0.72)q^{-2}. \quad (17)$$

With the chosen desired poles p_1 and p_2 , the desired characteristic polynomial of equation (7), corresponding

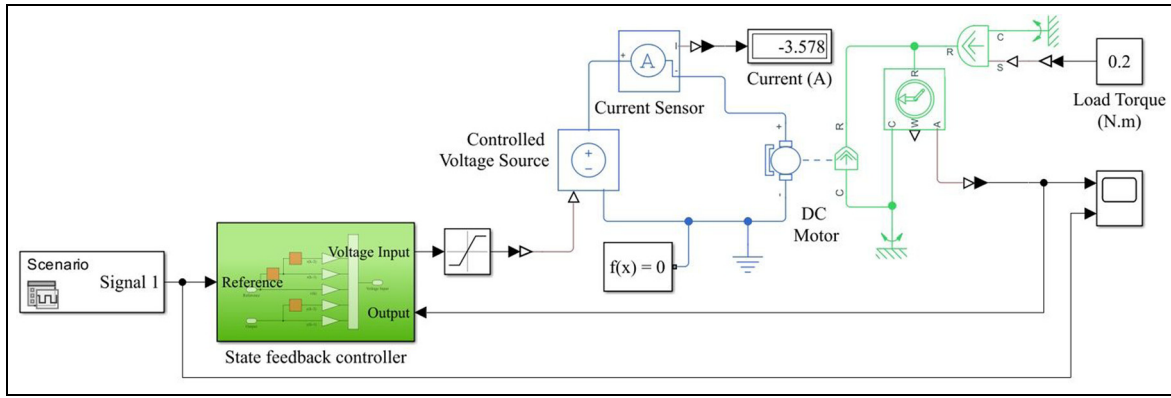


Figure 3. Schematic diagram of state feedback tracking control for DC motor.

to state feedback approach, is represented in the following form

$$\Delta_{sf}^d(q^{-1}) = (1 - p_1 q^{-1})(1 - p_2 q^{-1}) = 1 - (p_1 + p_2) q^{-1} + p_1 p_2 q^{-2}. \quad (18)$$

By matching the coefficients of q^{-1} and q^{-2} in equations (17) and (18), the coefficients k_1 and k_2 holds the following constraint

$$k_1 = 34.4 - 20(p_1 + p_2), k_2 = 20p_1 p_2 - 14.4. \quad (19)$$

The control parameter $G_{sf}(q^{-1}) = g_o + g_1 q^{-1} + g_2 q^{-2}$ is then determined by

$$g_o = 20, g_1 = -20(p_1 + p_2), g_2 = 20p_1 p_2. \quad (20)$$

Consequently, the state feedback tracking control input is introduced as

$$u_{sf}(k) = -k_1 y(k-1) - k_2 y(k-2) + g_o r(k) + g_1 r(k-1) + g_2 r(k-2). \quad (21)$$

The reference input is generated by a multi-step function using the Scenario/Signal 1 block, with the step amplitude of $\{\pi/6; 0; -\pi/6\}$ during the simulation time of 3 s. The state feedback controller manipulates the voltage supply for the DC motor through a saturation function, ensuring the voltage stays within practical physical limitations. It should be noted that the modeling of a motor driver is omitted, with the gain set to 1. Figure 3 illustrates the schematic diagram of state

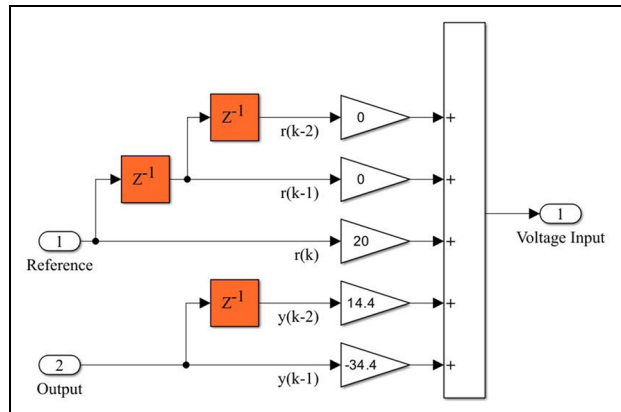


Figure 4. State feedback controller structure.

feedback tracking control, while its structure is shown in Figure 4 for the given DC motor described in Section ‘Model Identification using RLS’.

For the sake of comparison, there are three pairs of pole values, ranging from -0.5 to 0.5 , are sequentially chosen to design the desired characteristic of the closed loop tracking control system, as shown in Table 2. The simulation time is set to 3 s, while the sampling time is set to 0.01 s, expressed in delay blocks. The simulation results of the state feedback tracking controller with various pole values are depicted in Figure 5. The dashed black line denotes the reference input, while the system outputs corresponding to the desired poles $\{p_1, p_2\}$ of $\{0, 0\}$, $\{0.5, 0.5\}$, $\{-0.5, -0.5\}$ are expressed in solid red, solid blue, and solid green lines, respectively. It can be seen that the system outputs are stable at the steady state for all cases, but they oscillate during the transient

Table 2. Polynomial coefficients of state feedback control parameter gains.

Pole (p_1)	Pole (p_2)	k_1	k_2	g_o	g_1	g_2
0	0	34.4	-14.4	20	0	0
0.5	0.5	14.4	-9.4	20	-20	5
-0.5	-0.5	54.4	-9.4	20	20	5

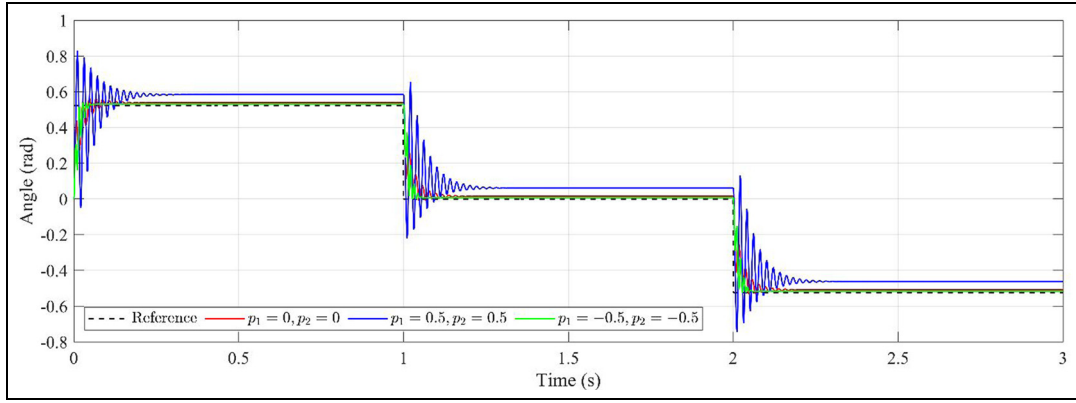


Figure 5. Simulation results of the state feedback tracking controller.

phase. Specifically, the system output corresponding to the pole of $\{0.5, 0.5\}$ exhibits the worst performance, with high overshoot and large transient time. In contrast, the output for the poles of $\{-0.5, -0.5\}$ performs better, displaying low overshoot and short transient time.

State feedback control with integral action

State feedback tracking control with integral action

It is worth noticing that the state feedback control exhibits a large steady-state error due to the mismatch between identified parameters and practical parameters, as well as the substantial assumption that $y_{ss} = r(k)$ which leads to the lack of robustness of tracking control. The augmentation system using the integration of the tracking error is necessary to eliminate the steady state error, which is described as follows:

$$(1 - q^{-1})v(k) = y(k) - r(k). \quad (22)$$

A state feedback tracking control input with integral action based on equation (22) is introduced as

$$u_a(k) = -K_a(q^{-1})y(k) - G_a(q^{-1})v(k), \quad (23)$$

where the subscript “a” denotes elements belonging to the control approach with integral action; $K_a(q^{-1})$ of degree α , and $G_a(q^{-1})$ of degree β , are polynomials of control parameter gains.

By using the state feedback tracking control input with integral action of equation (23), the tracking problem for the system of equation (1) can be represented as

$$(B(q^{-1})K_a(q^{-1}) + A(q^{-1}))y(k) = -B(q^{-1})G_a(q^{-1})v(k). \quad (24)$$

Substituting equation (24) into equation (22), it yields

$$\begin{aligned} & ((B(q^{-1})K_a(q^{-1}) + A(q^{-1}))(1 - q^{-1}) \\ & + B(q^{-1})G_a(q^{-1}))v(k) \\ & = -(A(q^{-1}) + B(q^{-1})K_a(q^{-1}))r(k). \end{aligned} \quad (25)$$

It can be seen that the stability of the closed loop control system of equation (25) ensures that $v(k) \rightarrow 0$, i.e. $y(k) \rightarrow r(k)$ in the steady state. To this end, the desired characteristic polynomial of equation (7) must be consistent with the characteristic polynomial $\Delta_a(q^{-1})$ of degree n_Δ , which is defined as follows:

$$\Delta_a(q^{-1}) = (A(q^{-1}) + B(q^{-1})K_a(q^{-1}))(1 - q^{-1}) + B(q^{-1})G_a(q^{-1}). \quad (26)$$

The polynomials of control parameter gains $K_a(q^{-1})$ and $G_a(q^{-1})$ are determined by matching the coefficients of the backward shift operators in the two characteristics polynomial equations.

Design example of state feedback tracking control with integral action

This section presents the control design of the ARX model of equation (16), with the polynomials of control parameter gains chosen as $K_a(q^{-1}) = k_{a1}q^{-1} + k_{a2}q^{-2}$, $G_a(q^{-1}) = g_a q^{-1}$.

Consequently, the characteristic polynomial of equation (26) is derived as

$$\begin{aligned} \Delta_a(q^{-1}) &= 1 + (0.05k_{a1} - 2.72 + 0.05g_a)q^{-1} \\ &+ (2.44 + 0.05k_{a2} - 0.05k_{a1})q^{-2} \\ &- (0.72 + 0.05k_{a2})q^{-3}. \end{aligned} \quad (27)$$

From equation (27), it is evident that the characteristic polynomial $\Delta_a(q^{-1})$ is of degree 3; thus, the desired characteristic polynomial $\Delta_a^d(q^{-1})$ features three poles p_1, p_2 , and p_3 , which are

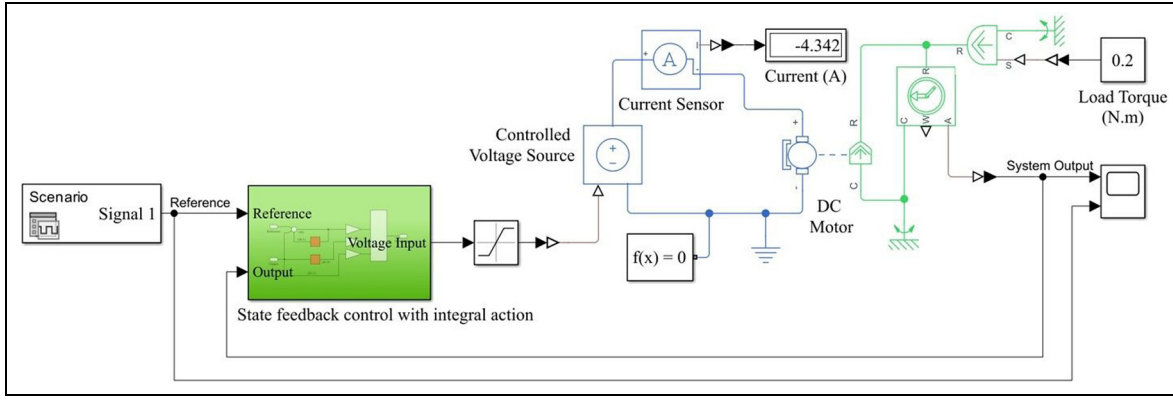


Figure 6. Schematic diagram of state feedback tracking control with integral action for DC motor.

$$\begin{aligned}\Delta_a^d(q^{-1}) &= (1 - p_1 q^{-1})(1 - p_2 q^{-1})(1 - p_3 q^{-1}) \\ &= 1 - (p_1 + p_2 + p_3)q^{-1} \\ &\quad + (p_1 p_2 + p_1 p_3 + p_2 p_3)q^{-2} - p_1 p_2 p_3 q^{-3}.\end{aligned}\quad (28)$$

Equating equations (27) and (28), the polynomials of control parameter gains hold the following constraints

$$\begin{cases} 0.05 k_{a1} - 2.72 + 0.05 g_a = -(p_1 + p_2 + p_3) \\ 2.44 + 0.05 k_{a2} - 0.05 k_{a1} = p_1 p_2 + p_1 p_3 + p_2 p_3 \\ 0.72 + 0.05 k_{a2} = p_1 p_2 p_3 \end{cases}\quad (29)$$

The simulation setup is consistent with the previous section, with the step amplitude of the reference input set to $\{\pi/6; 0; -\pi/6\}$ during the simulation time of 3 s. Figure 6 illustrates the schematic diagram of state feedback tracking control with integral action, while its structure is shown in Figure 7 for the given DC motor described in section ‘Model Identification using RLS’.

It can be seen from equation (29) that the state feedback control with integral action is characterized by three poles. Therefore, the first two poles are chosen as in the previous section, while the third pole is chosen as -0.5 and 0.5 , forming six pole pairs listed in Table 3. The system output under state feedback control with integral action is shown in Figure 8. For all pole pairs, the system outputs closely track the reference input in the steady state. For the case b), corresponding to the pole pair of $\{p_1, p_2, p_3\} = \{0.5, 0.5, 0.5\}$, the system output exhibits the smallest overshoot but the longest rising time. In

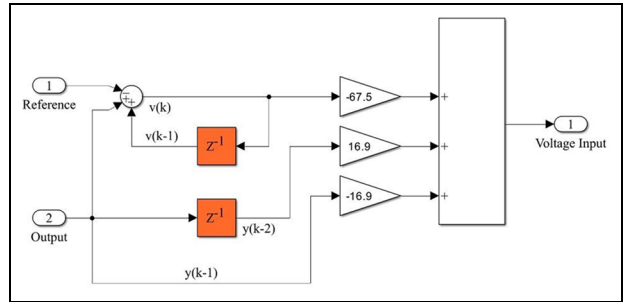


Figure 7. State feedback with integral action controller structure.

contrast, cases a) and f), corresponding to the pole pairs of $\{p_1, p_2, p_3\} = \{0, 0, 0.5\}$ and $\{-0.5, -0.5, -0.5\}$, performs better with small overshoot and short rising time. For cases c), d) and e), the system outputs exhibit oscillatory behavior, indicating underdamped control systems, with high-frequency vibrations during the transient phase. It should be noted that the poles for these cases are located on both left and right planes. It can be observed that the system output has a longer response time when the poles p_1, p_2 changes from $\{p_1 = 0, p_2 = 0\}$ to $\{p_1 = 0.5, p_2 = 0.5\}$. Comparing the cases b) and e), the system output is more fluctuating when the pole p_3 changes from 0.5 to -0.5 .

Mean deviation coupling strategy for synchronous control of steering wheels

Mean deviation coupling strategy

The synchronization error for each steering wheel is calculated as the difference between its steering angle

Table 3. Polynomial coefficients of state feedback control parameter gain with integral action.

Case	Pole (p_1)	Pole (p_2)	Pole (p_3)	k_{a1}	k_{a2}	g_a
a	0	0	0.5	34.4	-14.4	10
b	0.5	0.5	0.5	21.9	-11.9	2.5
c	-0.5	-0.5	0.5	41.9	-11.9	22.5
d	0	0	-0.5	34.4	-14.4	30
e	0.5	0.5	-0.5	36.9	-16.9	7.5
f	-0.5	-0.5	-0.5	16.9	-16.9	67.5

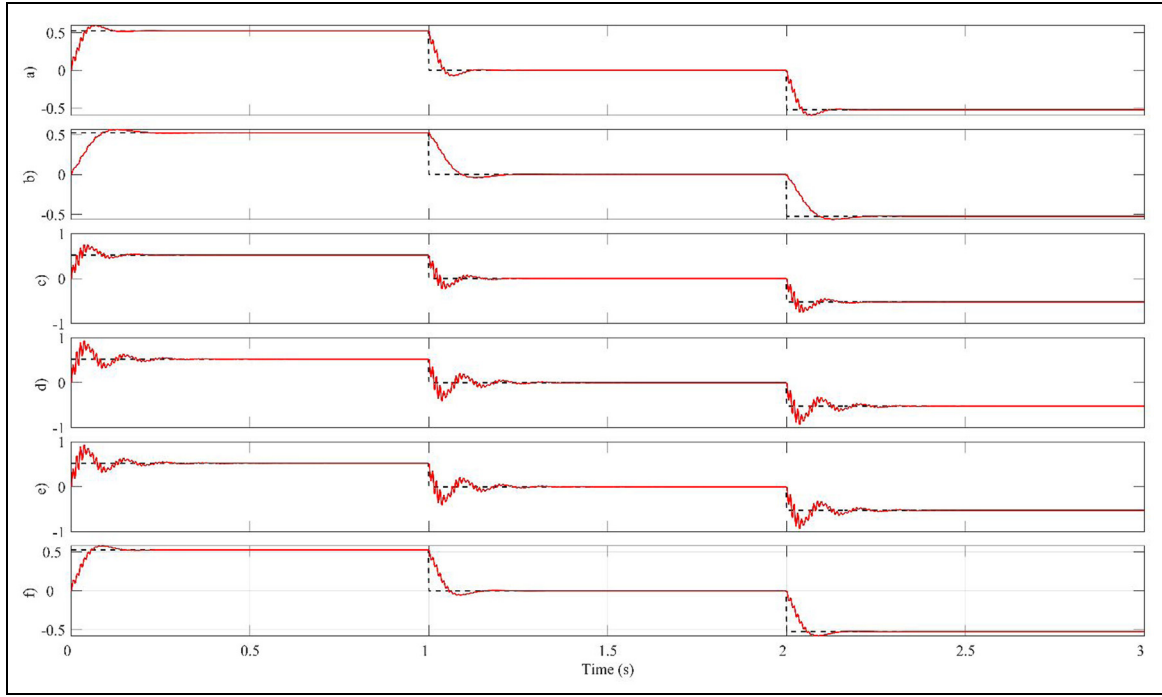


Figure 8. Simulation results of state feedback tracking controller with integral action.

and the average steering angle of the remaining wheels, expressed as:

$$es_i = \frac{1}{n} \sum_{j=1}^n y_j(k) - y_i(k), \quad (30)$$

where es_i and y_i are respectively the synchronization error and the steering angle of the i -th steering wheel, n is the total number of steering wheels of the MR.

The simultaneous position deviation of each steering wheel causes es_i to be non-zero, generating a compensating control signal at the motor input. The control input, based on the synchronization error, is then determined as follows:

$$u_i(k) = -K_i(q^{-1})y_i(k) - G_i(q^{-1})v_i(k) + C_i es_i(k), \quad (31)$$

where C_i is the compensation gain of the i -th steering wheel.

Despite the inclusion of the compensating control signal, the state feedback composition of the control system remains consistent with equation (25). Therefore, the characteristic polynomial $\Delta_a(q^{-1})$ of degree n_Δ of each steering wheel is determined similarly to equation (26), and the polynomials of control parameter gains are defined as in Section ‘State feedback tracking control with integral action’.

Design example of mean deviation coupling strategy

Suppose an MR model requires synchronous steering control for motor pairs, each comprising a left and right

Table 4. Setting parameters of two Simscape DC motor blocks in synchronous control simulation.

Setting	Left motor	Right motor
Armature inductance	0.0308 mH	0.0952 mH
Stall torque	15,700 mN.m	13,200 mN.m
No-load speed	4070 rpm	2125 rpm
No-load current	1030 mA	1986 mA

Table 5. The ARX model of two motors in synchronous control simulation.

Setting	ARX model
Left motor	$y_1(k) = 1.72 y_1(k-1) - 0.72 y_1(k-2) + 0.05 u_1(k)$
Right motor	$y_2(k) = 1.69 y_2(k-1) - 0.69 y_2(k-2) + 0.03 u_2(k)$

motor. The system models of the two DC motors are derived similarly to Section ‘Case study: system identification of a DC motor’ with the parameters described in Table 4.

Each motor is of permanent magnet type with a rated DC supply voltage of 24V. The ARX model and characteristic polynomial of each DC motor are determined similarly to Section ‘Case Study: System Identification of a DC Motor’, with the results presented in Table 5.

The ARX model structure for both motors are identical; therefore, their characteristic equations are derived from equation (26) and expressed in the general form as follows:

$$\Delta_{ai}(q^{-1}) = 1 + (-a_{i1} - 1 + b_i k_{ai1} + b_i g_{ai})q^{-1} + (a_{i1} - a_{i2} + b_i k_{ai2} - b_i k_{ai1})q^{-2} - (b_i k_{ai2} - a_{i2})q^{-3}, \quad (32)$$

where a_{i1}, a_{i2} and b_i are coefficients of the ARX model $\Delta_{ai}(q^{-1})$ of i -th motor. The polynomials of control parameter gains are determined by matching the coefficients of the backward shift operators in equations (28) and (32). The parameters are described in Table 6 with pole pairs of $\{p_1, p_2, p_3\} = \{0.5, 0.5, 0.5\}$.

Figure 9 shows the schematic diagram of the two-motor synchronous position control strategy based on MDCS and SFTC with integral action. Specifically, the compensation signal of each motor is formulated by the product of the compensation gain and the difference between the average position of the two motors and its

Table 6. Polynomial coefficients of SFTC parameter gains with integral action of two motors in synchronous control simulation.

Motor	k_{a1}	k_{a2}	g_a
Left	21.9	-11.9	2.5
Right	35.5	-18.83	4.17

current position. Each motor is controlled by SFTC with integral action, the structure of which is described in Figure 7.

Figure 10 illustrates the schematic diagram of the simulation of the external force components acting on each motor. This simulation aims to evaluate the synchronous motion behavior of two motors when one is subjected to disturbances. Specifically, it examines two scenarios: one where the motor experiences a sudden force at a specific moment (Figure 10(a)) and another where it is affected by noise over a period of time (Figure 10(b)). In the scenario in Figure 10(a), a force applied to the left motor causes a 10-degree deviation from its equilibrium point at $t = 0.6$ s. In the scenario depicted in Figure 10(b), the disturbance affecting the right motor is generated using a random function following $\sim \mathcal{N}(0, 1)$ with a sampling time of 0.01 s, applied at $t = 1.2$ s.

Figure 11 presents the simulation results for position tracking control of two wheels, comparing cases with/without the synchronization strategy. The evaluation focuses on three key moments: startup ($t = 0$ s), external force impact on the left motor ($t = 0.6$ s), and disturbance on the right motor ($t = 1.2$ s). At the initial stage, the angular position error of the left and right

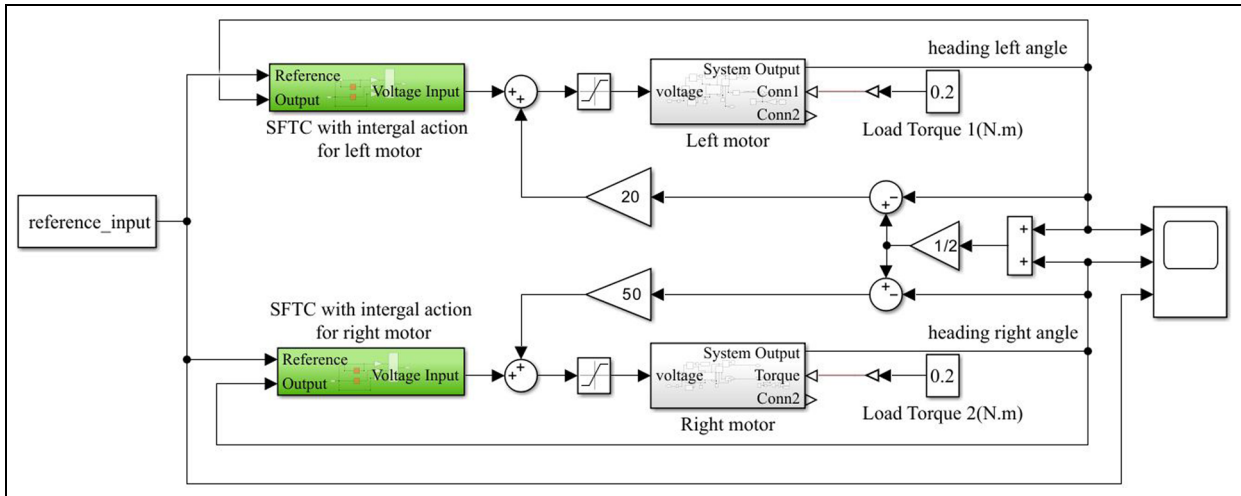


Figure 9. Schematic diagram of two-motor synchronous position control strategy using MDCS and SFTC with integral action.

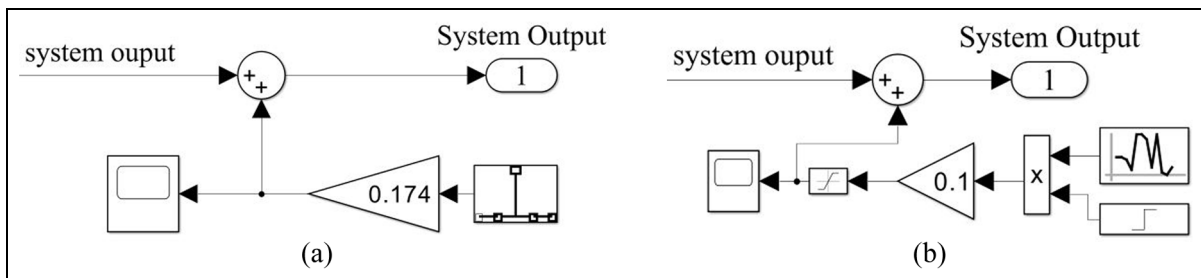


Figure 10. Schematic diagram of the simulation of external forces acting on the steering angle of left motor (a) and disturbances affecting right motor (b).

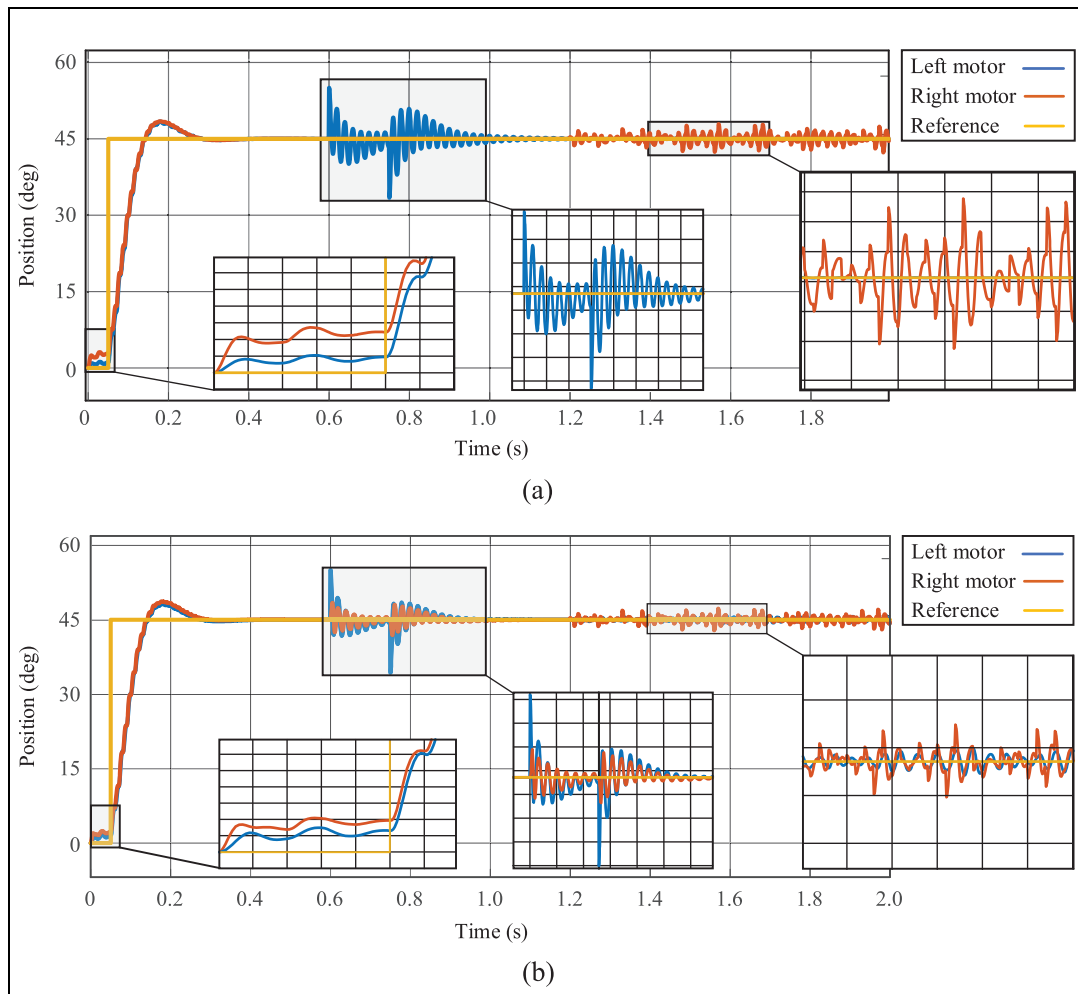


Figure 11. The positions of two wheels in the case of (a) without and (b) with the synchronous control strategy.

motors in the case without a synchronous strategy is greater in the absence of a synchronization strategy than in its presence. Furthermore, under synchronous control, both motors exhibit a similar position change pattern, which becomes evident from $t = 0.05$ s onward. This confirms the effectiveness of the synchronization control strategy from the moment both motors start operating. However, the deviation between each motor's position and its reference shows only a slight change before and after applying the control strategy. Specifically, the deviation in the right motor decreases, while that in the left motor increases.

At $t = 0.6$ s, the external force causes a significant increase in the tracking error of the left motor, which then gradually returns to its equilibrium position (45°) after 0.6 s in the asynchronous case and 0.4 s in the synchronous case. This demonstrates that applying MDCS improves the system's settling time. Additionally, in the synchronous case, the right motor exhibits oscillatory behavior that closely follows the left motor, a characteristic absent in the asynchronous case. These results highlight the effectiveness of the synchronization

strategy when either motor experiences a significant disturbance.

At $t = 1.2$ s, the angular position of the right motor starts to oscillate with small amplitude and high frequency due to the disturbance. In the synchronous case, this disturbance induces oscillations in the left motor with reduced amplitude and frequency, a phenomenon absent in the asynchronous case. Compared to the scenario at $t = 0.6$ s, the dynamic responses of the two motors exhibit divergence, except during periods of low oscillation frequency, such as between $1.5 - 1.6$ s and $1.8 - 1.9$ s. This suggests that MDCS struggles to achieve full synchronization, particularly under significant frequency variations. Notably, disturbances in one motor also affect the other, albeit with lower amplitude and frequency.

An additional simulation scenario for the synchronous control structure based on mean deviation coupling strategy combining proportional-integral-derivative (PID) controller²⁴ is performed with the aim of peer evaluation with the proposed algorithm. Specifically, each motor is regulated by a separate PID

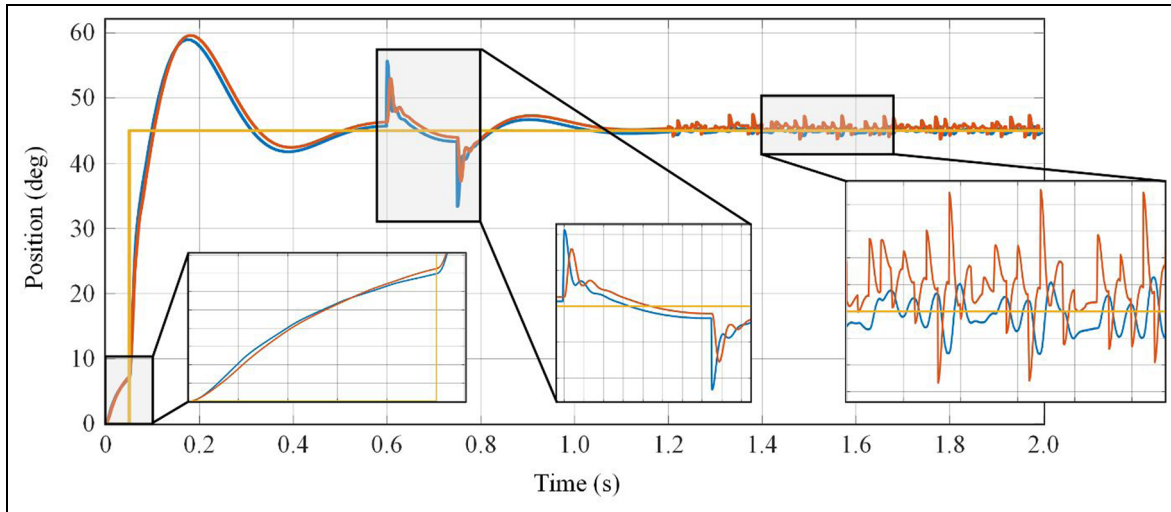


Figure 12. The position of the wheels is controlled by a PID controller combined with MDCS.

controller, while the control signal is supplemented with a compensation component from the MDCS. The adjustment parameters in the PID controller are tuned based on the Ziegler-Nichols theory²⁵ (desired settling time is about 0.3 s), specifically $[K_p, K_i, K_d]$ of the left and right motor are $[1.893, 30.867, 0.009]$ and $[0.141, 5.021, 0.049]$ respectively. Besides, the gain of MDCS in this simulation scenario is set similar to the proposed algorithm for the purpose of fairness in evaluation. Figure 12 illustrates the simulation results of the system under the PID-MDCS controller. Regarding the synchronization effect, the motors tend to follow each other's motion with a deviation of no more than 2.5° (except during major disturbances at 0.6 s). Compared to PID-MDSC, the proposed method achieves superior synchronization, with deviation between motors remaining below 1.5° (except during major disturbances). In the case of motors subjected to disturbances, the synchronization effect at the two motors is not maintained under the control of the PID-MDCS algorithm, which is contrary to the proposed algorithm. For each motor response, the PID controller causes a larger overshoot percentage (33.33%) than the SFTC with integral action (8.33%). In addition, the graph shows that the motors require more than 1 s to reach asymptotic steady state, while the proposed algorithm requires only 0.3 s. This is due to the influence of the magnitude of the proportional and integral terms; However, reducing their values to improve the overshoot results in a large settling time. Furthermore, the responses of the two motors after the transient time (1 s) do not converge completely to the reference angle. This proves that the PID-MDCS cannot control the steady-state error, which is not the case in the proposed algorithm. In summary, SFTC with integral action combined with MDCS shows better performance than PID-MDCS in terms of both the response behavior of each motor (settling time, overshoot, steady-state error) and the synchronization effect.

Practical application

System identification of permanent magnet steering wheel

Figure 13 illustrates the experimental test bench configured based on the structure of a VTR, comprising steering wheel assemblies and an electrical box responsible for system control. It is important to note that this study considers motion synchronization in pairs of steering wheels (left and right wheels) rather than all four wheels simultaneously. Each steering wheel assembly consists of a DC motor directly mounted to the steering axle and a potentiometer for measuring the steering angle.

In the experimental setup of this study, the ARX model for each wheel follows equation (1), incorporating a second-order $A(q^{-1})$ polynomial and a zero-order $B(q^{-1})$ polynomial. The RLS algorithm is set up such that the forgetting factor is 0.995, and the initial covariance matrix with the large positive numbers is $10^{20,26}$. The motors at each steering wheel are supplied with a fixed voltage of 7.2 V, and their steering angles are recorded using a potentiometer with a sampling time of 0.05 s. The ARX model coefficients are estimated over a 12-second period, with the resulting parameters

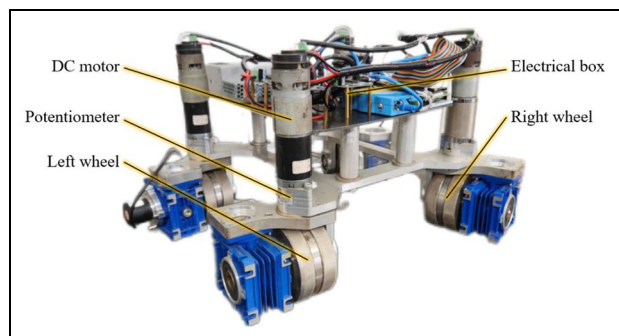


Figure 13. Experimental system of two-wheel steering VTR.

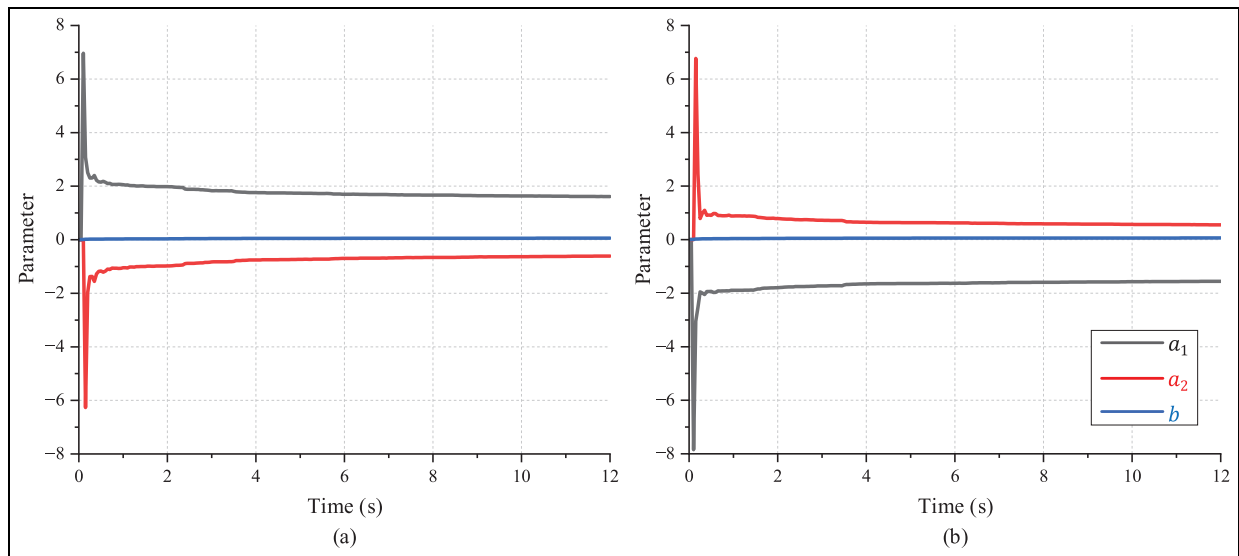


Figure 14. The coefficients of the ARX model for (a) left wheel and (b) right wheel.

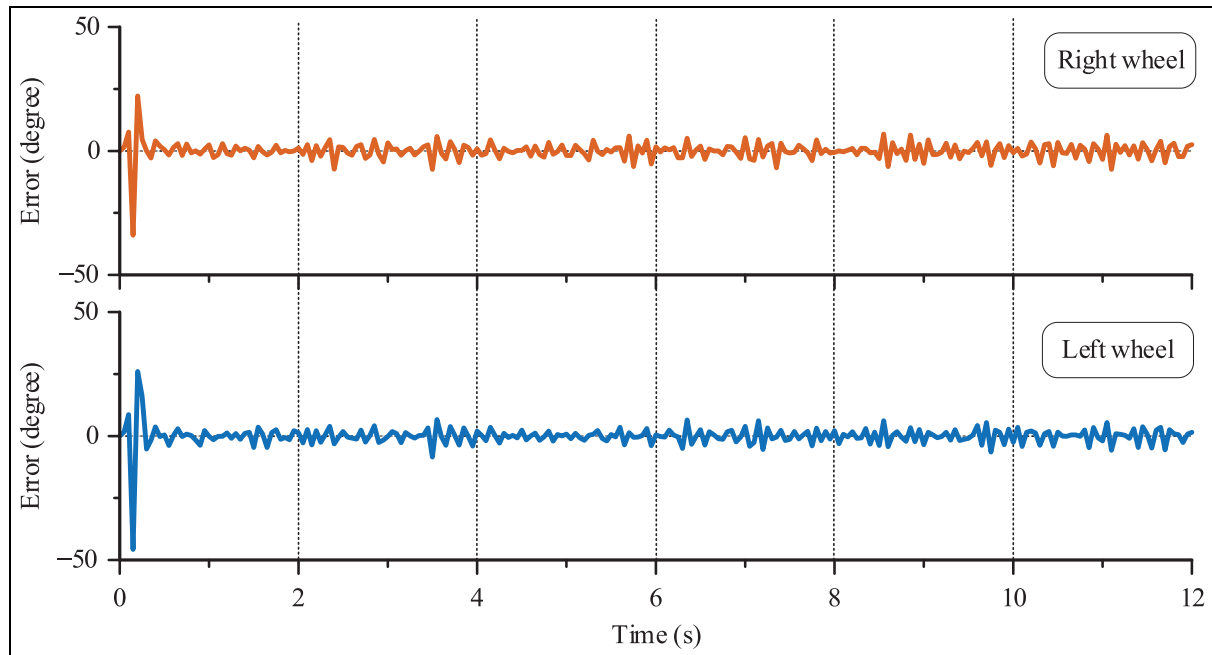


Figure 15. The estimated errors of the ARX model for left wheel and right wheel.

presented in Figure 14 and the estimated errors illustrated in Figure 15.

Figure 14 indicates that the coefficients tend to converge to specific values, with similar variations observed when considering both wheels. In particular, the coefficients of $A(q^{-1})$ polynomial change more significantly than $B(q^{-1})$ polynomial. The coefficient b in the left and right wheel models gradually increased but remained small, rising by 0.04 and 0.03, respectively, in the first second, followed by an increment of 0.01 per second until the fourth second, after which the increase slowed noticeably. In contrast, the coefficients a_1 and a_2 exhibited significant variation, progressing through

three distinct stages: initialization, convergence, and stabilization.

The initialization phase occurs between $t = 0$ to 0.15 s, during which the coefficients a_1 and a_2 undergo sharp decreases and increases, respectively. Specifically, the coefficients $[a_1, a_2]$ of left and right wheels are $[-7.844, 6.770]$ and $[6.963, -6.265]$, respectively. This sudden change is attributed to the large positive values, which lead to unreliable initial parameters and significant fluctuations during the update process.²⁰

Furthermore, the estimated error from Figure 15 indicates that the initialization phase exhibits a large error, rendering the ARX model inaccurate during this

Table 7. The ARX model of two wheels in VTR.

Setting	ARX model
Left wheel	$y_1(k) = -1.448y_1(k-1) + 0.448y_1(k-2) + 0.066u_1(k)$
Right wheel	$y_2(k) = 1.490y_2(k-1) - 0.490y_2(k-2) + 0.063u_2(k)$

Table 8. The parameter gains of SFTC with integral action of two wheels.

Wheel	k_{a1}	k_{a2}	g_a
Left	-31.409	8.682	1.894
Right	13.730	-5.794	1.984

period. The convergence phase occurs between $t = 0.15$ to 10 s, during which the coefficients of polynomial $A(q^{-1})$ gradually stabilize. Notably, the rapid convergence of a_1 and a_2 takes place between 0.15 to 0.225 s. Beyond this point, the coefficients $[a_1, a_2]$ change more gradually, ultimately converging to $[-1.571, 0.569]$ for left wheel and $[1.635, -0.633]$ for right wheel. According to Figure 15, the estimated error in this stage approaches and remains near zero, indicating that the ARX model is reliable. Compared to the period of large overshoot values, the RLS algorithm requires only approximately 0.075 s to update the values close to the convergence point. This rapid convergence is attributed to the large initial diagonal values of the covariance matrix, which accelerate the algorithm's convergence speed. Additionally, a large forgetting factor enhances algorithm stability but slows convergence due to the influence of regression. This effect is evident in Figure 14, where the convergence slope gradually decreases, indicating a slower rate of coefficient change. The stability phase begins after 10 s, at which point the model coefficients remain nearly constant. The final converged coefficients $[a_1, a_2, b]$ for the left and right wheels are $[-1.448, 0.448, 0.066]$ and $[1.490, -0.490, 0.063]$, respectively. These coefficients are used in this study for controller design. The final ARX models for both wheels are determined and presented in Table 7. Based on this identified model, the controller gain parameters are calculated using the approach described in Sections 'State feedback control with integral action' and 'Mean deviation coupling strategy for synchronous control of steering wheels', with the selected poles set to $\{0.5, 0.5, 0.5\}$, as detailed in Table 8.

Mean deviation coupling strategy and state feedback tracking control with integral action of synchronous control of steering wheel

The mean deviation coupling strategy was implemented with gain coefficients of 3.0 for the left wheel and 3.5 for the right wheel. The experiment was conducted over

a 10-second duration, featuring two desired angle scenarios: 45° (from 0 to 4.12 s) and 90° (from 4.12 to 10 s). The sampling time was set to 0.05 s, with an initial steering angle of 0° for both wheels.

Figure 16 and Figure 17 depict the steering angles of the left and right wheels under non-synchronous and synchronous control strategies, respectively. The analysis focuses on key operational phases: startup (zone A), desired angle transition (zone B), and steady-state operation (zone C).

In zone A, the steering angles of both wheels increase sharply at 0.15 s, reaching 60° (left wheel) and 103° (right wheel) in the synchronous case, compared to approximately 70° for both wheels in the asynchronous case. The settling time is 0.5 s without the synchronization strategy and 0.7 s with it. Additionally, significant oscillations occur in the synchronous case during the first 0.5 s. The results indicate that in zone A, the synchronous control strategy has drawbacks in terms of overshoot and settling time. However, the experimental results also show that the left and right wheels exhibit similar movement patterns. This is because the compensation signal from the synchronous control strategy is significantly larger than the control signal from SFTC, altering the system dynamics and causing the eigenvalues to deviate from their intended placement. Compared to the simulation results, the inability to maintain the designed pole positions, combined with actual disturbances, significantly impacts steering angle behavior. However, reducing the compensation gain also decreases the system's synchronization efficiency, presenting a trade-off between stability and synchronization performance.

In zone B, the absence of MDCS, such as Figure 16, results in noticeable discrepancies between the steering angles of the two wheels. Specifically, the left wheel demonstrates high control performance, achieving a rapid response time of 0.1 s with no overshoot. In contrast, the right wheel exhibits significant overshoot (125°) and a delayed response of 0.6 s. This discrepancy arises because the selected pole placement $\{0.5, 0.5, 0.5\}$ are not optimal for the model of the right wheel. At $t = 4.2$ s, the steering angles of the two wheels differ by a maximum of 35.45° before gradually converging over the next 0.6 s. This instability in steering control prevents the MR from establishing an instantaneous center of velocity. In the case of MDCS, as shown in Figure 17, the left wheel tends to follow the right wheel with a smaller amplitude, consistent with the observations in Section 'Design example of mean deviation coupling strategy'. However, the left wheel exhibits greater overshoot than in the asynchronous case, reaching 140° at $t = 4.2$ s. This suggests that while synchronization occurs, the overall errors in both wheels increase. Additionally, synchronous control introduces oscillations in both wheels, a phenomenon absent in other cases. This instability stems from the failure to maintain the designated pole placement, as previously observed in zone A. Notably, the difference in steering angles between the two wheels remains similar in both cases,

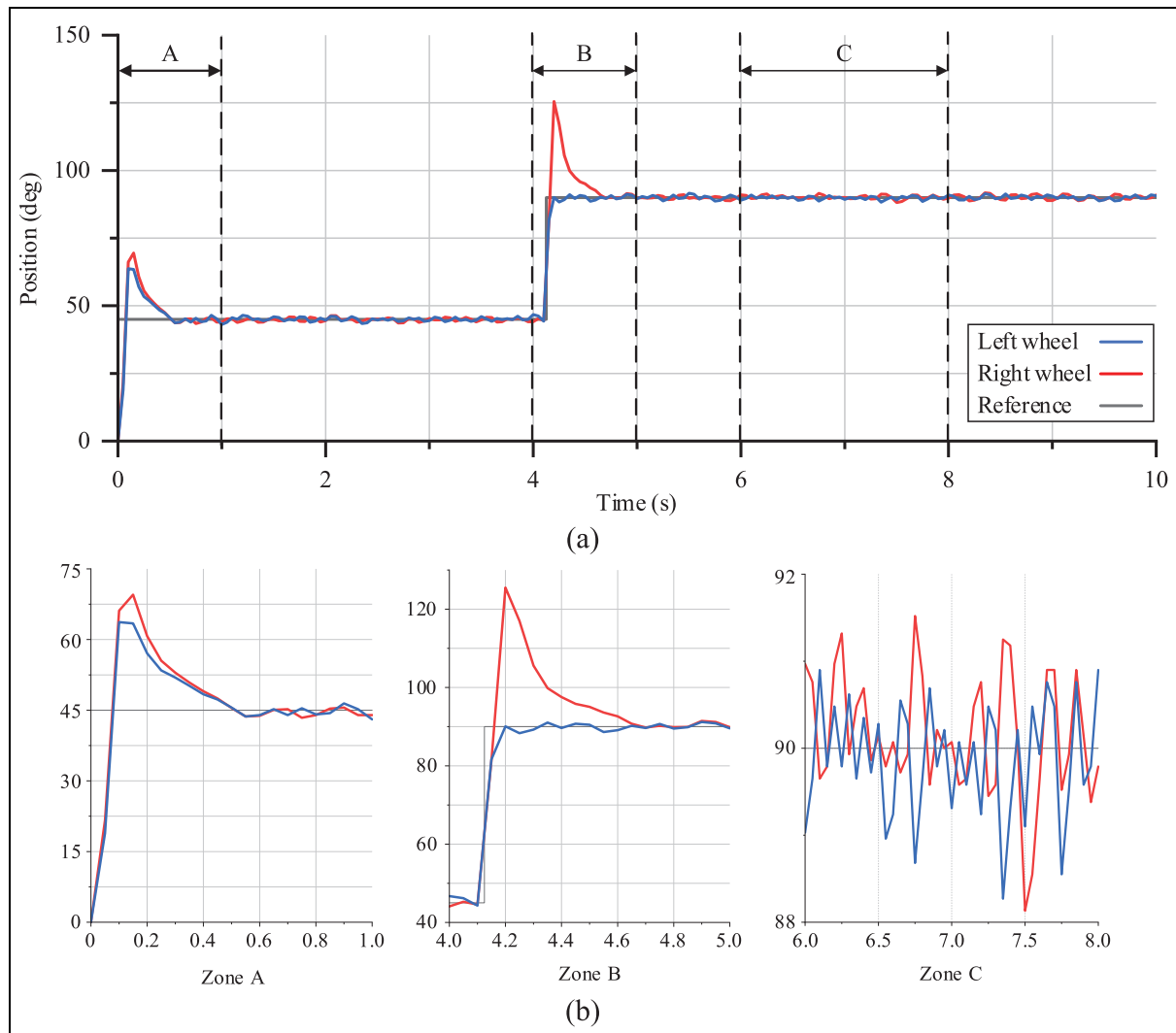


Figure 16. Wheel positions without the synchronous control strategy: (a) overall trends and (b) zone-specific trends.

approximately 40° . This indicates that synchronization is achieved, but does not fully resolve the issue of establishing an instantaneous center of velocity for the MR. Furthermore, the settling time in the synchronous case (0.4s) is shorter than in the asynchronous case (0.6s), aligning with the discussion in Section ‘Design example of mean deviation coupling strategy’.

In zone C, the variation in steering angles closely aligns with the simulation results for the disturbance scenario. Specifically, the steering angle synchronization effect is absent during high-frequency oscillations (above 10Hz, as indicated by the recorded trend) but occurs at lower frequencies. The position trend from 6.75 to 7.2s provides evidence of this synchronization effect. Additionally, the oscillation amplitude of both wheels in the synchronous case is larger than in the asynchronous case, consistent with observations in the other zones.

In summary, the experimental results indicate that MDCS does not demonstrate superiority over independent control under the given conditions. However, this observation is specific to the selected pole placement $\{0.5, 0.5, 0.5\}$ for SFTC with integral action for both

wheels and the chosen MDCS gain parameters of 3.0 for the left wheel controller and 3.5 for the right wheel controller. While the results are insufficient to confirm the feasibility of MDCS for VTR applications, the experiment successfully validates the operational integration of SFTC with integral action and MDCS.

Conclusion

The study proposed a method for synchronous position control of steering wheels by integrating MDCS and SFTC with integral action, optimizing computational efficiency. Besides, this paper systematically presents the design process of SFTC controller with integral action and demonstrates the feasibility of the algorithms by simulation. This establishes a foundation for further research on state feedback-based control methods. The simulation results in each controller design section are specifically implemented, evaluating the behavior of the control system at theory-based design poles. This is an important suggestion in zoning the values of the poles in optimization problems. Moreover, the simulation results also demonstrate the feasibility of

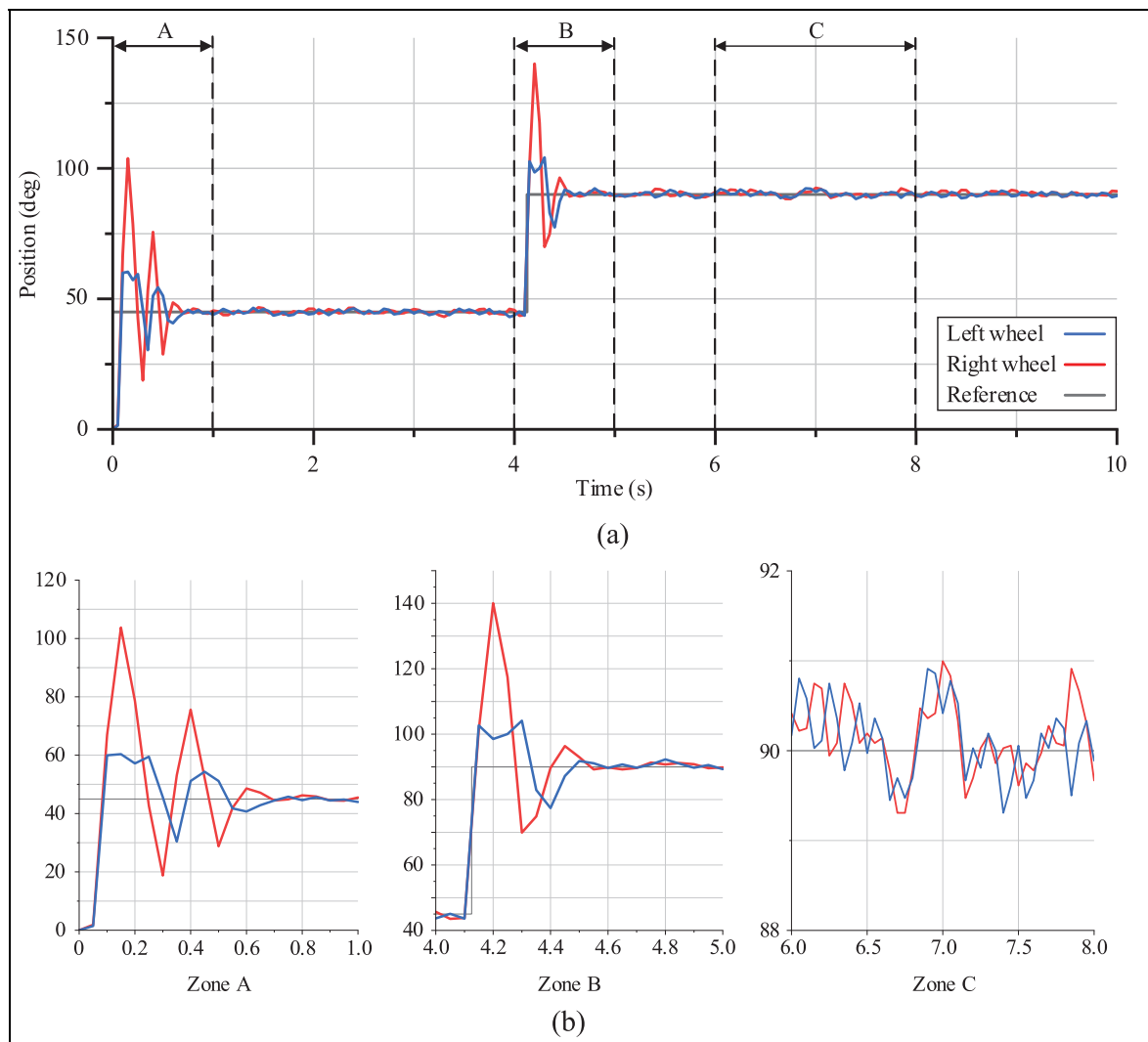


Figure 17. Wheel positions with the synchronous control strategy: (a) overall trends and (b) zone-specific trends.

synchronous control, along with some important observations when applying MDCS are summarized as follows:

- Simulation results demonstrate the potential reduction in settling time when applying the proposed algorithm.
- Simulation and experimental results reveal that the algorithm struggles to achieve synchronous motion control for high-frequency oscillations.
- In applications where the proposed algorithm synchronizes multiple subsystems, a disturbance affecting one subsystem propagates to the others with reduced amplitude and frequency.

A limitation of this study is the absence of a method for selecting optimal pole placements and appropriate MDCS gain coefficients. While the experimental results demonstrate the effect of synchronous control, it is insufficient to confirm the algorithm's suitability for VTR applications. Additionally, the advantages observed in

the simulation results are not fully reflected in the experimental outcomes. In terms of computational volume, the proposed method requires a complex initial computational process including determining the desired poles and calculating the gains. Besides, the proposed controller design has not considered the disturbance component, leading to no basis to affirm that the algorithm can be applied robustly in practical conditions.

Building on these limitations, the next phase of this research will focus on developing a method for determining optimal compensation gains and pole placements for each system. The objective is to mitigate unnecessary oscillations during the overshoot phase, enhance convergence time, and maintain synchronization effectiveness. Additionally, increasing the number of wheels in synchronous control experiments is essential to comprehensively evaluate the robustness of the proposed method. For the disturbance component, a disturbance observer combined with the proposed algorithm needs to be designed to ensure robustness in the control strategy.

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Ethical considerations

This article does not contain any studies with human or animal participants.

There are no human participants in this article and informed consent is not required.

Author contributions

Nhut Thang Le: Writing – original draft, Software, Investigation. Cong Toai Truong: Visualization, Formal analysis, Data curation. Huy Hung Nguyen: Validation, Resources, Project administration. Tan Tien Nguyen: Formal analysis, Conceptualization, Funding acquisition. Van Tu Duong: Writing – review & editing, Supervision, Methodology.

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Data availability statement

Data will be made available on reasonable request.

References

1. Taraglio S, Chiesa S, De Vito S, et al. Robots for the energy transition: A review. *Processes* 2024; 12: 1–35. Epub ahead of print 1 September 2024. DOI: 10.3390/pr12091982
2. Rosa GL, Michele M, Giovanni M, et al. A low-cost lightweight climbing robot for the inspection of vertical surfaces. *Mechatronics* 2002; 12: 71–96.
3. Love P, Kalra J, Gu J, et al. A wall climbing robot for oil tank inspection. In: *2006 IEEE International Conference on Robotics and Biomimetics*. China: IEEE, 2007, pp.1523–1528.
4. Daniel S and Karsten B. Climbing robots for maintenance and inspections of vertical structures: a survey of design aspects and technologies. *Robotics and Autonomous Systems* 2013; 61: 1288–1305.
5. Le NT, Truong CT, Nguyen HH, et al. Yaw angle determination of a mobile robot operating on an inclined plane using accelerometer and gyroscope. *Measurement* 2025; 247: 1–13. Epub ahead of print 15 April 2025. DOI: 10.1016/j.measurement.2025.116806
6. Changhui S and Weicheng C. Review of underwater ship hull cleaning technologies. *Journal of Marine Science and Application* 2020; 19: 415–429.
7. Pei Y, Jidong J, Lingyu S, et al. Design and analysis of a push shovel-type hull-cleaning wall-climbing robot. *Journal of Field Robotics* 2024; 42: 1569–1585. Epub ahead of print 4 September 2024. DOI: 10.1002/rob.22430
8. Yuki U, Takashi O, Kazuhiko T, et al. The development of driving system with differential drive steering system for omni-directional mobile robot. In: *2009 IEEE International Conference on Mechatronics and Automation*. Changchun: IEEE, 2009. Epub ahead of print 18 September 2009. DOI: 10.1109/ICMA.2009.5246142
9. Jiwei Q, Hongji L, Zhe Z, et al. Performance analysis and optimization for steering motion mode switching of an agricultural four-wheel-steering mobile robot. *Agronomy* 2022; 12: 1–17. Epub ahead of print 1 November 2022. DOI: 10.3390/agronomy12112655
10. Chun YL, He W, Xin D, et al. Multi-motor position synchronization control method based on non-singular fast terminal sliding mode control. *PLoS One* 2023; 18: 1–33. Epub ahead of print 1 June 2023. DOI: 10.1371/journal.pone.0281721
11. Alfian M, Adha IC, Samiadji H, et al. Tracking control of high order input reference using integrals state feedback and coefficient diagram method tuning. *IEEE Access* 2020; 8: 182731–182741.
12. Ahmed S, Abdallah H, Ashraf H, et al. Quadcopter trajectory tracking control using state-feedback control with integral action. *International Journal of Computer Applications* 2017; 168: 975–8887.
13. Xin D and Guang HY. An adaptive approach to state feedback tracking control of systems with actuator failures. In: *2007 American Control Conference*. New York: IEEE, 2007, pp. 3997–4002.
14. Uiliam NLTA, Ricardo B, Luiz EP, et al. Discrete-time modeling and state feedback control of a DC motor with inertial load. *Journal of Applied Research and Technology* 2022; 20: 718–728.
15. Niu B, Wang D, Alotaibi ND, et al. Adaptive neural state-feedback tracking control of stochastic nonlinear switched systems: An average dwell-time method. *IEEE Transactions on Neural Networks and Learning Systems* 2019; 30: 1076–1087.
16. Jue Y, Jian Z and Dehong Y. State feedback integral control for a rotary direct drive servo valve using a Lyapunov function approach. *ISA Transactions* 2015; 54: 207–217.
17. Adlane B, Dominique K and Eric O. State feedback control with full or partial integral action for large scale winding systems. In: *Fourtieth IAS Annual Meeting. Conference Record of the 2005 Industry Applications Conference*. IEEE, 2005. Epub ahead of print 6 October 2005. DOI: 10.1109/IAS.2005.1518469.
18. Lebao L, Lingling S and Shengzhou Z. Mean deviation coupling synchronous control for multiple motors via second-order adaptive sliding mode control. *ISA Transactions* 2016; 62: 222–235.
19. Karl JA and Richard MM. *Feedback systems: an introduction for scientists and engineers*. Princeton: Princeton University Press, 2009.
20. Le NT, Nguyen MT, Truong CT, et al. Recursive least squares algorithm for online parameter identification of DC motor: Theory and practice. In: *10th International*

- Conference on Engineering and Emerging Technologies (ICEET)*. IEEE, pp.27–28.
21. Lennart L. Perspectives on system identification. *Annual Review of Control, Robotics, and Autonomous Systems* 2010; 34: 1.
 22. Qiao L, Xue Y, Gui Y, et al. Comparative study of non-linear black-box modeling for power electronics converters. *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, Nashville, TN, USA, 2023, pp. 1446–1452, doi: 10.1109/ECCE53617.2023.10362794.
 23. Duong VT, Truong CT, Nguyen TT, et al. State space model identification using model reference adaptive approach: Software and hardware-in-the-loop simulation. *Cogent Engineering* 2024; 11: 1–13. Epub ahead of print 2024. DOI: 10.1080/23311916.2024.2434938
 24. Mu Y, Qi L, Sun M, et al. An improved deviation coupling control method for speed synchronization of multi-motor systems. *Applied Sciences* 2024; 14: 1–14. Epub ahead of print 1 June 2024. DOI: 10.3390/app14125300
 25. Åström KJ and Hägglund T. Revisiting the Ziegler-Nichols step response method for PID control. *Journal of Process Control* 2004; 14: 635–650.
 26. Yassine K. Recursive identification of induction motor parameters. *Simulation Modelling Practice and Theory* 2004; 12: 363–381.